COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE12S4
DATE PERFORMED:	October 9, 2026	**DATE SUBMITTED:**	October 9, 2026
NAME:	Cardenas, Lea Marie De Vera, Sophia Hare, Andrei Angelo Pangyarihan, Jenny Rose Penaflorida, Wrio Carlo Prieto, Irish Crystle	**INSTRUCTOR:**	Engr. Jimlord Quejado

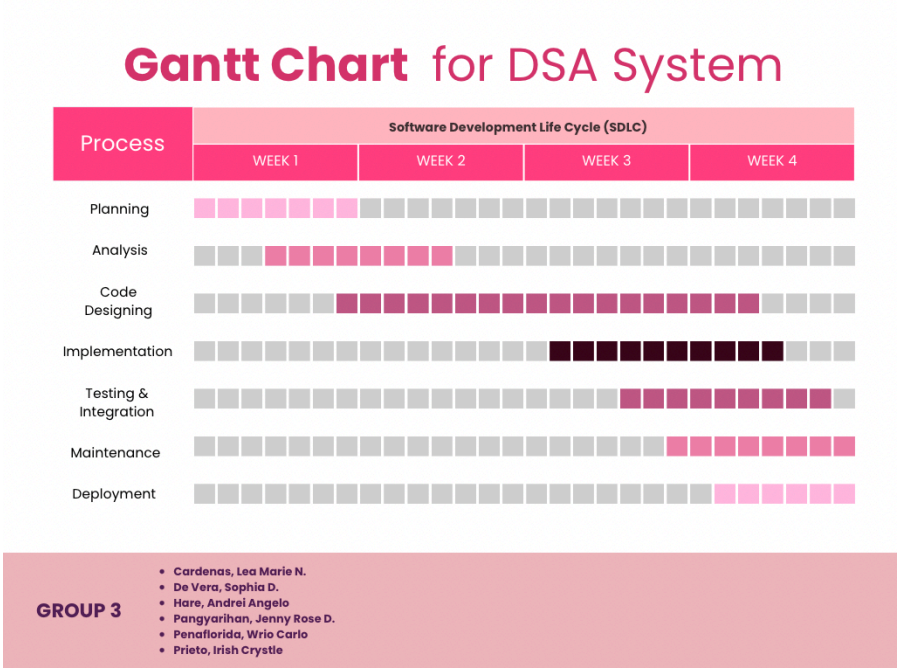
SEATWORK 11.1 PLANNING

1. PROCEDURE OUTPUT

- ☑ ~~Form your group for the project and submit the names in a 1/4 sheet of yellow paper.~~
- Write down your 3 proposed systems to develop.

SYSTEMS PROPOSED	FUNCTIONS / PURPOSE
Contact Booklist	- The system's purpose is "supposed to be connected" to our phone book and/or viber and emails, where the contact numbers, landline numbers, and emails are directly listed here. The system is to help people track down the people's contact details and emails, greatly helpful in several ways. - O(1) to read the most relevant contact instantly, and O(log n) to update the pool structure when a new contact is inserted or a contact profile is modified.
Network Routing specifically tracking the DHCP Leasing Time	- The system will greatly aid the system administrators to track the IP addresses. Its primary purpose is to help network administrators manage, allocate, and audit IP addresses across complex corporate topologies in real time. The system stores available IP pools inside a min-heap. The lowest available IP address is always near the root. - O(1) to read the next available IP, and O(log n) to update the pool structure when an address is extracted or returned.
Inventory Management System	- The system's purpose is to sort and make inventory processes smooth and to manage and monitor products that include product names, prices, available stock, and sales transactions. It helps users track the number of products available, record incoming and outgoing stocks, and identify items that need to be restocked.

- Meet with your group and decide what type of SDLC you will use to create a gantt chart and discuss the task responsibilities.



CANVA LINK: https://canva.link/31raey62frmvn1u

1. **Planning** → this stage is where we'll plan the project and brainstorm the possible system we will be implementing.
2. **Analysis** → this stage is where we will analyze what type of searching algorithm techniques, trees, and graphing will be used in the system.
3. **Code Designing** → this stage is where we design the program's structure: flowcharts and pseudocode for each searching algorithm. We planned to make the code design long to make better adjustments accordingly.
4. **Implementation** → this is where we will implement the program
5. **Testing and Integration** → this is the stage where we'll try to check if its working
6. **Maintenance** → the debugging stage especially when we check whether there are anomalies and hallucinations in the program.
7. **Deployment** → trying to deploy the program to a small population to check if the code is working outside of our group.

- What possible 10 questions will you ask to find out how the business process works for one of your proposed systems.

Contact Booklist

1. What happens when someone has two contacts with the same name?
2. How do you edit or delete a contact when a number changes or is no longer used?
3. How should the system handle the number that is wrong, blocked and not to be reachable?
4. Who will use the contact booklist, and what do they mainly use it for?
5. How many people will use the system at the same time?
6. How should the system show the results when a search matches many contacts?
7. What device and internet conditions will the system run on?
8. What happens if you face the same current way of keeping contacts?
9. Do you need to attach extra details to a contact
10. Do users' names need to log in and and should different users have different permissions?

Network Routing specifically tracking the DHCP Leasing Time

1. Where is DHCP handled in your network (router, dedicated server, or both), and who looks after it?
2. How long are leases set to last, and does that change depending on the device type or network area (staff, guests, Wi-Fi, servers)?
3. What was the reasoning behind those lease durations, and when did anyone last revisit them?
4. How do you keep track of which devices have an address, and when each lease begins and ends?
5. What takes place when available addresses are nearly used up, and how do you get alerted?
6. Which devices need a fixed or reserved address (printers, servers, CCTV), and where are those reservations documented?
7. How are renewals, expired leases, address conflicts, and unauthorized DHCP servers dealt with?
8. How frequently do users or devices join and leave the network (students, visitors, phones), and does this influence the lease length?
9. Which reports or notifications would you want, like active leases, usage per subnet, or soon-to-expire leases, and who should receive them?
10. How long are lease logs kept, where are they stored, and are they used for audits or security checks?

Inventory Management System

1. How do you currently record and look up your items (SKU, barcode, notebook, spreadsheet)?
2. How do customers or staff usually search for a product, and what problems do they run into (similar names, misspellings, not knowing the exact name)?
3. How are your products organized (categories, shelves, aisles), and how often do items get moved or reclassified?
4. What happens when stock comes in or goes out, and how do you fix a wrong entry?
5. How do you know when an item is running low, and who decides when to reorder?
6. Which products have an expiry date or shelf life (paint, glue, sealant), and how do you decide which batch to sell first?

7. What sales or stock reports do you need (daily, weekly, by date range), and how do you prepare them now?
8. Which items sell fastest or get asked for most often, and how do you identify them?
9. How do you decide what to reorder when your budget is limited?
10. What are the biggest problems with your current process (wrong counts, lost items, slow searching, duplicates), and what would you want the new system to fix first?

☑ ~~Submit answers in a pdf file format.~~

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google Overview**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Help in idea brainstorming, concept helping and lesson clarification**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.